Christian Domínguez

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chdominguez

Skills

Programming Languages: Python, HTML, CSS, SASS, JavaScript, TypeScript, C, Swift, Shell (bash, zsh)
Frameworks and Tools: Flask, FastAPI, Pydantic, Dash, React, Next.js, Vite, WebPack, ParcelJS, MongoDB, SQLite, SQLAlchemy, PostgreSQL, Keycloak, LDAP, OAuth2, NGINX, Caddy, Docker, Docker Swarm, Git, CI/CD, AWS, Nextflow, Supabase, Vercel, Celery, Redis
Machine Learning: Transformers, Hugging Face, Pandas, Scikit-learn, Feature Engineering, Model Training

Experience

CEO / CTO *Barcelona, Spain*
Temple Compute *Feb 2026 – Present*

github.com/temple-compute | horus-runtime

- Co-founded Temple Compute, building Horus, a runtime and workflow orchestration platform for executing HPC workflows, spanning bioinformatics and broader scientific computing, with LLM-driven workflow scaffolding and flexible compute environments.
- Leading both technical strategy and business development, spanning product architecture, engineering execution, investor relations, and early team hiring.
- Driving core platform development, including `horus-runtime` async subprocess streaming and the `tc-os` backend (FastAPI, SQLModel/PostgreSQL, Redis, MinIO).

Software Engineering Lead *Barcelona, Spain*
Nostrum Biodiscovery *Jan 2026 – Oct 2026*

- Team lead for software engineering, owning architecture decisions, code quality, and delivery cadence across concurrent platform projects.
- Continued development of the Nostrum Suite plugin ecosystem and web application, while directing technical priorities for the engineering team.

Full Stack Developer, Platform Engineer *Barcelona, Spain*
Nostrum Biodiscovery *2023 – Dec 2025*

- Worked as platform engineer, engineering custom scientific applications for both internal users and external collaborators across diverse bioinformatics and computational biology domains.
- Architected and deployed production-grade web applications handling complex scientific workflows, including molecular dynamics databases, bioinformatics pipelines, and machine learning prediction tools.
- Engineered cross-platform desktop applications utilizing modern web technologies, ensuring functionality in local and offline environments.
- Developed and maintained CI/CD pipelines using self-hosted GitLab CI/CD, streamlining deployment processes for multiple concurrent projects.
- Key projects and technologies:
  - **VSE Vall d'Hebron Enterovirus Repository:** Built full-stack bioinformatics platform integrating Keycloak authentication, Nextflow pipeline orchestration, Next.js frontend, PostgreSQL database with FastAPI backend, and Celery task queue with Redis for asynchronous job processing.
  - **Nostrum Suite Plugin Development:** Engineered modular plugin architecture using Flask backend with Pydantic for validation, integrated with Vite + React + TypeScript frontend for molecular simulation configuration and execution.
  - **Nostrum Suite WebApp:** Built browser-based interface for launching and managing molecular simulations with Flask backend, SQLite database, and React frontend.
- **Technology Stack:** Flask, FastAPI, Pydantic, ReactJS, Next.js, TypeScript, Vite, ParcelJS, WebPack, SQLite, MongoDB, PostgreSQL, Docker, GitLab CI/CD.

Visitor *Barcelona, Spain*
Barcelona Supercomputing Center (BSC) *Jan 2026 – Present*

- Continued affiliation with BSC projects as an External Visitor.

Research Engineer II *Barcelona, Spain*
Barcelona Supercomputing Center (BSC) *2025 – Dec 2025*

- Continued as platform engineer on HPC and bioinformatics tooling, in dual affiliation with Nostrum Biodiscovery.
- Key projects and technologies:
  - **MDPosit:** Architected molecular dynamics database on AWS infrastructure using MongoDB and Docker Swarm for high-availability deployment.
  - **PredIG:** Developed interpretable T-cell epitope immunogenicity predictor as a Horus plugin, implementing frontend interfaces and Docker containerization with Pandas and Transformers integration.
  - **aMLProt:** Created comprehensive machine learning pipeline plugin for feature extraction, selection, outlier detection, and model training, with interactive table visualization frontend.
  - **YieldDB:** Developed web-based platform for collecting and analyzing protein expression data using FastAPI backend, SQLite database, and React frontend.
- **Technology Stack:** FastAPI, Pydantic, Next.js, PostgreSQL, MongoDB, Keycloak, Nextflow, Celery, Redis, Docker, Docker Swarm, AWS, Transformers, Pandas.

Research Engineer I *Barcelona, Spain*
Barcelona Supercomputing Center (BSC) *2023 – 2024*

- Joined BSC as platform engineer, in dual affiliation with Nostrum Biodiscovery, laying the technical foundation for the Horus HPC workflow platform.
- Key projects and technologies:
  - **Horus Workflow Manager:** Architected comprehensive HPC and bioinformatics workflow management platform with plugin ecosystem. Developed core workflow engine with React frontend and Flask backend using SQLite. Built plugin repository portal using Next.js, MongoDB, and FastAPI for community plugin sharing.
  - **Horus-OpenAuth-LDAP Integration:** Implemented enterprise authentication system integrating LDAP as source of truth, Keycloak as Identity Provider for OIDC/OAuth2 tokens, oauth2-proxy for request authentication, Caddy reverse proxy for authentication enforcement, and custom provisioning middleware for automatic SQLite user creation.
  - **Mol* Contributions:** Contributed to open-source molecular visualization library, implementing features for enhanced structure rendering and user interaction capabilities.
- **Technology Stack:** Flask, FastAPI, React, Next.js, SQLite, MongoDB, Keycloak, LDAP, OAuth2, Caddy, oauth2-proxy.

Researcher *Barcelona, Spain*
Universitat de Barcelona *2022-2023*

- Simulated catalytic surfaces for CO_2 conversion in energy-efficient reaction environments.
- Developed `ParAMS AutoTrain`, a Python-based automation tool to accelerate parameter fitting workflows and enhance model training.
- Improved result throughput and reproducibility via robust automation and HPC integration.

Researcher *Barcelona, Spain*
Universitat Autònoma de Barcelona *2021*

- Modeled interstellar chemical reaction pathways using quantum chemical simulations on HPC clusters.
- Automated job submission and post-processing using Bash, Python, and SLURM scripting, significantly reducing time-to-results.

Education

Barcelona Supercomputing Center *2023 – Present*
Internal Training Programs

- High-Performance Computing (HPC) fundamentals and optimization techniques
- Software development best practices and design patterns
- Advanced Docker containerization and orchestration
- CI/CD pipeline development and deployment strategies

Udemy *2023*
Web Development Certifications
- HTML/CSS, JavaScript, React, and SQL.
- Quantum computing.
- Swift & SwiftUI.

Erasmus Mundus Joint Master Degree *2021 – 2023*
MSc in Theoretical Chemistry and Computational Modelling (TCCM)
- Focused on quantum chemistry, molecular simulations, and high-performance computing.
- Studied at multiple European universities as part of the Erasmus Mundus consortium.
- Completed relevant coursework at the University of Perugia in:
  - Web deployments using GitHub and Heroku
  - Machine Learning and AI tools for scientific applications
  - HPC development with OpenMPI and GPU programming
- Developed a Hartree–Fock algorithm to compute the quantum energy of the hydrogen molecule at the University of Salamanca.

Universitat Autònoma de Barcelona *2017 – 2021*
BSc in Chemistry
- Solid foundation in physical chemistry and computational methods.
- Participated in undergraduate research in theoretical modeling.

Publications

Horus: A reproducible workflow manager for bioinformatics 2025
C. Domínguez-Dalmases, A. Cañellas-Solé, R. Lambea-Jané, M. Municoy, V. Guallar

10.1101/2025.09.26.678751 — *BioRxiv*

Water formation on interstellar silicates: role of Fe^{2+}/H_2 interactions in the $O + H_2 \rightarrow H_2O$ reaction 2021
C. Domínguez-Dalmases, Albert Rimola, Marc Serra-Peralta
10.1039/d2cp04051d — *Physical Chemistry Chemical Physics*
ISSN: 1463-9076 / 1463-9084

Other Projects

TipTapChat *AI Travel Planning Assistant*
- AI-powered travel itinerary planner with multi-version trip comparison, calendar integration, and mobile-responsive chat interface featuring drag-and-drop scheduling and external calendar synchronization
- Tools Used: TypeScript, React, Next.js, Vercel, Supabase, shadcn/ui

MedSum *AI Medical Report Summarizer*
- Privacy-preserving medical document summarization using local AI models for secure, offline processing of sensitive healthcare data
- Tools Used: FastAPI, Vite, React, TypeScript, Hugging Face Transformers

Atomic Molecular Visualizer
github.com/chdominguez/atomic-molecular-visualizer
- Render and manipulate atomic structures on macOS and iPadOS

- Tools Used: Swift, SwiftUI, SceneKit

MC Simulations *github.com/chdominguez/mc-simulations*
- Monte Carlo algorithms for simulating atomic liquids
- Tools Used: C

Charted! *apps.apple.com/app/charted-regression-charts*
- Create and view regression charts on iOS devices with interactive touch controls
- Tools Used: Swift, SwiftUI

Languages

Spanish, Catalan: Native
English: C1 (Advanced)
French: A2 (Elementary)
Chinese: HSK2 (Elementary)